Online Algebra, Logic and Topology Seminar

(CMUC, University of Coimbra)

Speaker: Hugo Luiz Mariano (Universidade de São Paulo (IME-USP), Brazil)
Date: December 1st, 2026
Time: 15:00 (Europe/Lisbon)
Place: Google Meet

On completeness conditions on quantale valued sets

(joint work with José Goudet Alvim and Caio de Andrade Mendes)

Abstract. Categories of quantale valued sets (Q-sets) has been proposed over the years in attempts to expand the celebrated notion of H-sets, for complete Heyting algebra H, to the broader category of parameter algebras consisting of certain quantales—those being (not necessarily commutative) residuated monoids in the category of posets and supremum-preserving maps.

In this work, we examine various notions of completeness for Q-sets (such as singleton completeness, gluing completeness, and net completeness), where Q is a commutative, semicartesian quantale—a structure that properly extends the notion of a locale and encompasses examples such as the quantale of ideals of a commutative ring with identity, as well as Lawvere's quantale $[0, \infty]$.